MLB Game Simulator: Implementation Plan

Team 23 · University of Michigan, Master of Applied Data Science capstone · November 2026

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Status. Data collection is implemented (notebooks/savant_scrape_updated.ipynb), though its latest changes and data/games/ are not yet committed. The simulator, models and results do not exist yet.

1. Question and test design

A game's outcome depends on many batter-pitcher matchups, so season records alone say little about a single game. We will simulate each game inning by inning, one plate appearance (a batter's turn at bat) at a time, about 1,000 times; a team's share of simulated wins is its win probability.

We develop on four seasons (2022–2025) and test on the fifth (2026). The 2026 regular season is already complete, so the test is retrospective: each 2026 game is predicted from 2022–2025 history plus that game's actual starting lineups and starting pitchers. 2026 scores, winners and full-season 2026 player totals are never prediction inputs and are read only after all predictions are made. Because the lineups and starters are known, the evaluation is **conditional on known starting lineups and starters**.

2. Data and prior work

Data. Play-by-play, box scores, starting lineups, schedules and roster moves come from the MLB Stats API; pitch-level data and season leaderboards come from Baseball Savant, with additional leaderboards from FanGraphs. The simulator's estimates come from the 2022–2025 play-by-play. We will write it in Python: pandas prepares the data, NumPy holds rates and draws seeded random outcomes, and SciPy fits the removal model and computes intervals (all already in requirements.txt).

Prior work. Markov-chain models (Bukiet and colleagues; Tango) build a game from plate-appearance outcomes and base-out states, so we simulate at that level. Log5 is a standard formula for combining batter and pitcher rates, and the Marcel projections motivate weighting recent seasons and pulling small samples toward the league average.

3. Simulator design

Items are agreed in our planning discussions unless marked **proposed**, which means they need team approval and the stated choice is our recommended default.

Part	Plan
Plate appearances	Multicategory log5 turns batter, pitcher and league rates (split by opponent handedness) into probabilities for 8 outcomes: strikeout, walk, hit by pitch, single, double, triple, home run, and in play with no hit.
Player rates	Short histories are blended toward league averages; players with no history get league averages. Season weighting (equal per plate appearance versus more weight on recent seasons) and blending strength are chosen in development.

Part	Plan
Runners	A historical table gives each play's end state, including where each runner ends up and any baserunning on the deciding pitch, from its outcome, bases and outs. Earlier steals and wild pitches come from a separate table (2023 onward, after that year's rule changes), so each movement, run and out counts once.
Starting pitcher	**Proposed:** a logistic regression, separate from log5 and fitted on earlier seasons, gives the chance of removal before each plate appearance from batters faced, simulated pitches, runs allowed, inning, outs, score margin and the pitcher's average length in earlier starts.
Bullpen	After the starter leaves, one pooled team profile pitches. It does not pick relievers or track workload, so rules such as the three-batter minimum are not represented. **Proposed:** leave out pitchers with uncertain roster status, with a sensitivity run that includes them.
Lineups and rules	Starters bat all game; designated hitter, extra-inning runner, walk-offs, no ties.

4. Implementation stages

Stages run in order in a new notebook (notebooks/game_simulator.ipynb) that writes no files. Owners are not yet assigned.

Stage and task	Needs	Output	Done when
1. Prepare inputs. Load game files, label outcomes, order games by date.	data/games	Clean tables	Each completed plate appearance has one outcome; counts match collection.
2. Player rates. Calculate rates for each weighting and blending option.	Stage 1	Rates as of 2024, 2025, 2026	Each set uses only earlier seasons; rates sum to 1.
3. Log5. Combine batter, pitcher and league rates.	Stage 2	Matchup probabilities	Average batter versus average pitcher returns league rates.
4. Runner tables. Build the whole-play and event tables.	Stage 1	Two transition tables	Replayed 2024 plays give a runs-per-inning table compared with actual.
5. Pitching. Fit the removal model; build bullpen profiles.	Stages 1–2	Removal model; two bullpens	Simulated starter length compared with actual 2024.
6. Game engine. Simulate complete games under the rules.	Stages 3–5	Simulated scores	Scripted test innings give expected results; no invalid states or ties.
7. Development evaluation. Compare options on 2024, then check 2025 once.	Stage 6	Chosen settings; 2024 and 2025 results	Settings fixed before the 2025 run.
8. 2026 backtest. Recalculate rates from 2022–2025; simulate each game 1,000 times.	Stage 7	Win probabilities, scores, results tables	All 2,430 games listed (1 cancelled); predictions done before results are joined.

Stages 1–6 show that the simulator runs correctly, not that its predictions are useful. Only the comparisons with baselines in Stages 7 and 8 can show that.

5. Evaluation and the final report

We will report winner accuracy (how often the team given over 50% won); log loss and Brier score, which penalize confident misses; calibration (whether teams given about 60% won about 60% of the time); and simulated against actual runs, margins and extra-inning frequency. Two baselines set the bar: the 2022–2025 home-team win rate (53%) for every game, and prior-season records combined by log5. Intervals come from resampling by date; simulation noise is at most about 1.6 points per game.

Report section	Content from this work
Introduction	The problem and the goal of estimating win probabilities by simulation (Section 1).
Methods	Prior work, data, the four-season development and fifth-season test, simulator design, Stage 7 settings.
Results	After Stage 8: metrics versus baselines, calibration chart, simulated versus actual runs, game coverage.
Discussion	Practical meaning; limitations and improvements; responsible use of player statistics; gambling risks.
Conclusion	What the 2026 results say about the original question, written only after Stage 8.

6. Limitations

- **Bullpen:** about 40% of team-games include an uncertain pitcher; such pitchers faced 4.3% of 2022–2025 relief batters.
- **Substitutes and newcomers:** non-starters (3.7% of plate appearances) are ignored; players without history get league averages.
- **2026 at game level only:** we chose not to collect 2026 play-by-play, though it exists.

Sources

Bukiet et al. (1997) · Tango Markov model · Log5 · Marcel projections · Designated hitter rule · Extra-inning runner · 2023 rule changes · Three-batter minimum · Brier score